+91 9875691340
Vellore, Tamil Nadu
gurumaujsatsangi@gmail.com

Gurumauj Satsangi

Portfolio: gurumaujsatsangi.in
github.com/GurumaujSatsangi
linkedin.com/in/gurumauj-satsangi

Experience

SDE Intern **May — July 2025**
Algoleap Technologies Pvt. Ltd. | *Certificate* *Hyderabad*

- Cut multimodal AI integration and rate-limit errors by 30% by architecting a unified GenAI API design spanning text, image, audio, and video generation, validated through unit and integration testing before deployment.
- Eliminated the need for 4 separate API keys by designing a single consistent GenAI interface.

Intern **Dec 2024 — Jan 2025**
Indian Association for the Cultivation of Science | *Certificate* | *Website* *Kolkata*

- Boosted departmental operational efficiency by 40% by architecting and shipping a login-secured Store Inventory Job Card Portal for end-to-end job card tracking.
- Processed 8,000+ applications with zero downtime by designing a scalable platform architecture.

Skills, Technologies and Certifications

Languages	HTML, CSS, JavaScript, Java, Python, SQL
Core CS Fundamentals	Object-Oriented Programming, Design Patterns, Operating Systems
Frameworks & Architecture	Node.js, Express.js, REST API Design, Unit Testing
Cloud and DevOps	AWS, Docker, CI/CD
Databases	MySQL, PostgreSQL, Redis, Neo4j, MongoDB
Certifications	AWS Certified Developer - Associate, AWS Certified Solutions Architect - Associate, AWS Certified Cloud Practitioner
Generative AI & Tools	LangGraph, OpenAI API, Prompt Engineering, Git/GitHub, Agile/Scrum, VS Code, GitHub Copilot

Projects

LiveLingo | Node.js, Express.js, AWS Interactive Video Service, Gemini Flash 3.1 Live **Github**

- Architected a fault-tolerant, low-latency streaming system handling continuous 20ms PCM audio chunks under constrained network conditions (350 Kbps), designing middleware to absorb TCP Head-of-Line blocking spikes (17.3s) without dropping the pipeline.
- Achieved sub-400ms end-to-end latency by engineering a WebSocket-based real-time data pipeline integrating AWS IVS with a translation service layer, evaluating output stability and semantic accuracy across languages.

Validex | *Node.js, Express.js, Azure Maps, Azure OCR, Twilio, Langgraph* **Github | Website | Demo Video**

- Architected a multi-agent healthcare data validation system, reducing manual verification workloads by 80-90% processing 100 provider records in under 5 minutes with 85%+ extraction accuracy from unstructured PDFs.
- Optimized validation pipeline lookup/matching logic using LangGraph-based workflow orchestration, cutting per-record processing time significantly at scale; Selected for EY Techathon 6.0 Semi Finale 2026 (Certificate)

Memory Lens | Flutter, Dart, Node.js, Express.js, AWS Rekognition, Qdrant DB, OpenAI Embedding **Github | Demo Video**

- Designed a scalable vector search architecture, achieving sub-2-second query latency across 1,000+ images with 85%+ semantic relevance.
- Built a voice-driven query interface using NLP-based embeddings, reaching 95%+ voice recognition accuracy; Selected for Samsung PRISM GenAI Hackathon Semi Finale 2025 (Certificate)

Conference Management Toolkit | Node.js, Express.js, Bootstrap, AWS EC2, RDS, Redis, Nodemailer **Github | Website**

- Designed a modular, role-based server architecture supporting 100+ concurrent users with 100% reliability, automating the review lifecycle and cutting manual admin processing time by 40

Publications

A System and Method for Voice-Driven Semantic Image Search Leveraging AWS and OpenAI, *V. Khurana, G. Satsangi, A. Sharma, Dr. Umamaheswari M - Accepted for presentation and publication, IEEE INCECT 2026 (Aug 21–22, 2026)*

Achievements

Honorable Mention Award, *Bangkok University x Berkley SkyDeck Fund Hackathon 2026* Certificate
Best AI Model, *Yantra Central Hackathon 2025, VIT Vellore* Certificate
Winner, *Aquathon 2024 (Technical Hackathon on World Water Day), VIT Vellore* Certificate

Education

Bachelor of Technology in Computer Science Engineering, *Vellore Institute of Technology, Vellore* 2023 - 2027
CGPA: 8.43 / 10.00